Full Four-Dimensional Variational Data Assimilation with a Learned Inverse Observation Operator under Observation Noise

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Abstract

Variational data assimilation (DA) estimates the initial state of a dynamical system by fitting a model trajectory to sparse observations, but when the observation operator is non-invertible the resulting cost function is riddled with poor local minima. Frerix et al. (2021) addressed this by training a learned inverse observation operator—a convolutional neural network (CNN) that lifts observations back into physical space—and using it to initialize and reshape the assimilation objective. We reimplement their approach in PyTorch and extend it in three ways: we replace their simplified observation-misfit objective with the full four-dimensional variational (4D-Var) cost, adding an explicit background term and observation-error covariance; we corrupt every experiment with additive Gaussian observation noise at three magnitudes; and we wrap the solver in an operational sliding-window (cycling) scheme for long-horizon assimilation. We evaluate on two systems: the one-dimensional (1D) Lorenz-96 model and two-dimensional (2D) Kolmogorov flow, an incompressible Navier–Stokes system. Across all noise levels, learned-inverse initialization combined with hybrid (physics-space then observation-space) optimization yields the lowest analysis root-mean-square error (RMSE), and sliding-window cycling stabilizes assimilation over horizons where single-window 4D-Var degrades. We also report honest negative results, including a background-error standard deviation whose tuning yielded only marginal gains and an unexpected ordering of its optimal value.

1 Introduction

Numerical weather prediction (NWP) launches a forecast by integrating a numerical model of the atmosphere forward from an estimate of its current state. Because the atmosphere is observed only incompletely, indirectly, and noisily, producing that initial estimate is itself an inverse problem: given a forecast model and a stream of imperfect measurements distributed in space and time, recover the system state well enough to forecast from it. This is the task of data assimilation (DA), and it is the component of an operational NWP system that most directly limits forecast skill. The flagship variational method is four-dimensional variational data assimilation (4D-Var), which estimates the initial condition $\boldsymbol{x}_0$ of an assimilation window by minimizing a cost that balances a prior ("background") estimate against the misfit between the model trajectory and the observations spread across that window.

DA is hard precisely because the systems of interest are chaotic: small errors in the analysis (the assimilated initial state) grow exponentially, so the analysis must be accurate and the optimization landscape is treacherous. The Lorenz-96 model is the canonical low-order testbed for exactly this difficulty, and it is the primary one-dimensional (1D) system we study here. In our configuration (grid size 40, forcing $F = 8$), trajectories started 10^{-6} apart separate at a fitted exponential rate $\lambda = 2.263\,/\mathrm{MTU}$, corresponding to a Lyapunov time $T_L = 1/\lambda = 0.442$ model time unit (MTU); this is somewhat shorter than the ∼0.6 MTU expected from the original study, a known artifact of our fixed-step integrator that we report honestly. Because errors saturate within roughly a Lyapunov time, an inaccurate analysis cannot be rescued downstream, and this is the regime in which a better treatment of the observations is supposed to help.

The learned inverse-observation idea. Frerix et al. [1] study the variational problem of recovering an initial state $\boldsymbol{x}_0$ whose model trajectory $\mathcal{M}^t(\boldsymbol{x}_0)$ fits a sequence of observations. The observation operator H in their setting (and ours) is a sparse spatial subsampling: it keeps every fourth grid point for Lorenz-96 (10 of 40 points observed) and every sixteenth point in each direction for two-dimensional (2D) Kolmogorov flow (a $4 \times 4 = 16$-point subset of 4096 grid points). Because H is many-to-one and highly lossy, the standard observation-space objective $\|H\,\mathcal{M}^t(\boldsymbol{x}_0) - \boldsymbol{y}_t\|^2$ is strongly non-convex and riddled with bad local minima, so a poorly initialized gradient-based optimizer gets trapped. Their contribution is to train a convolutional neural network (CNN) that acts as a learned inverse observation operator H_θ^{-1}, mapping an observation sequence back into physics (state) space. They use it in two complementary ways: as a better *initialization* (lift the observations to a physics-space first guess $H_\theta^{-1}(Y)$ instead of a naive interpolation or average), and as a *reformulation of the loss* (replace the observation-space misfit with a physics-space misfit $\sum_t \|\mathcal{M}^t(\boldsymbol{x}_0) - [H_\theta^{-1}(Y)]_t\|^2$, which removes the lossy H from inside the objective and is therefore much closer to convex). They further propose an optional *hybrid* schedule: run the smoother physics-space optimization first as a warm start, then refine against the true observations in observation space. Crucially for our work, the original study assimilates *clean* observations and uses a *simplified* objective (the observation-misfit term only, with no background term and no explicit observation-error covariance), and its reference implementation is in JAX/Flax.

Our contribution. This capstone tests whether the learned inverse-observation idea survives a more operationally realistic setting, through three extensions, all built on a from-scratch PyTorch reimplementation of the entire pipeline (the original is JAX/Flax)—the dynamical-system integrators, the CNN inverter, the correlation preconditioning, and the data-assimilation loop solved with the limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimizer. (i) *Full 4D-Var cost.* We replace the paper's observation-misfit-only objective with the complete 4D-Var cost $J(\boldsymbol{x}_0) = J_\mathrm{b} + J_\mathrm{o}$ (Eq. (3)), adding a background term with covariance $\boldsymbol{B} = \sigma_\mathrm{b}^2\boldsymbol{C}$ ($\boldsymbol{C}$ the system's spatial correlation) and an observation-error covariance $\boldsymbol{R} = \sigma_\mathrm{obs}^2\boldsymbol{I}$. The background term pulls the analysis toward a prior $\boldsymbol{x}_b$ and is what makes the problem well-posed when observations are sparse; σ_b is the background-error standard deviation we tune and σ_obs the observation-error standard deviation. (ii) *Observation noise.* Every experiment adds independent Gaussian observation noise with $\sigma_\mathrm{obs} \in \{0.1, 0.5, 1.0\}$ (the paper used clean observations), which both stress-tests robustness and probes a deliberate mismatch, since the inverter is trained at one noise level and evaluated at others. (iii) *Sliding-window cycling.* We add an operational sliding-window (cycling) 4D-Var extension in which short assimilation windows are advanced through time and each cycle's analysis is propagated forward to form the background for the next window—the way 4D-Var is actually run operationally, in contrast to the paper's single fixed window.

Our headline finding is that the paper's central claim broadly holds under these harder conditions: learned-inverse-observation initialization combined with hybrid optimization attains the lowest analysis root-mean-square error (RMSE) at every tested noise level, and hybrid optimization dramatically rescues a cheap climatological-mean baseline that observation-only optimization leaves stuck in a bad local minimum. We also report honest negative results, notably that a secondary expectation about how the optimal σ_b should shift did not hold, and that on the smoke-scale Kolmogorov-flow runs the cycling extension does not yet show a decisive advantage.

Roadmap. The remainder of the paper is organized as follows. Section 2 summarizes the learned inverse-observation approach of Frerix et al. [1] and states precisely how we extend it. Section 3 describes the models and methods: the Lorenz-96 and Kolmogorov-flow dynamical systems and their integrators, the observation-noise model, the full 4D-Var cost and its physics-space hybrid companion, the learned inverse observation operator, the decorrelation preconditioning, and the sliding-window cycling scheme. Section 4 presents results, first for Lorenz-96 (integrator diagnostics, inverter training, baseline initialization, the tuned-σ_b full 4D-Var sweep, and the sliding-window experiments) and then for two-dimensional Kolmogorov flow. Section 5 discusses limitations and caveats, separating what reproduces the paper from what our extensions add, and concludes.

2 Background: The Learned Inverse-Observation Approach and Our Extensions

2.1 The Method We Extend

DA estimates the state of a dynamical system from a forecast model together with a stream of incomplete, indirect, and noisy observations, so that a forecast can be launched from the resulting analysis. The flagship operational method is 4D-Var, which recovers the initial condition $\boldsymbol{x}_0$ of a forecast window by minimizing a cost that balances a prior estimate against the misfit between the model trajectory and observations distributed in time. The difficulty is intrinsic to the chaotic systems that DA targets: in the Lorenz-96 testbed small errors grow exponentially, so the analysis must be accurate and the optimization landscape is treacherous.

Frerix et al. [1] study the variational problem of recovering an initial state $\boldsymbol{x}_0$ whose model trajectory $\mathcal{M}^t(\boldsymbol{x}_0)$ fits a sequence of observations. The observation operator H is a sparse spatial subsampling: it retains every fourth grid point for Lorenz-96 (10 of 40 sites observed) and every sixteenth point in each direction for 2D Kolmogorov flow (a $4 \times 4 = 16$-site subset of 4096 grid points). Because H is many-to-one and highly lossy, the standard observation-space objective $\|H\,\mathcal{M}^t(\boldsymbol{x}_0) - \boldsymbol{y}_t\|^2$ is strongly non-convex and riddled with bad local minima, and a gradient-based optimizer started from a poor first guess is easily trapped.

The contribution of Frerix et al. [1] is to train a CNN that acts as a learned inverse observation operator H_θ^{-1}, mapping an observation sequence Y back into physical-state space. They exploit this operator in two complementary ways. First, as a better *initialization*: the observations are lifted to a physics-space first guess $H_\theta^{-1}(Y)$ instead of a naive interpolation or averaging baseline. Second, as a *reformulation of the loss*: the lossy observation-space misfit is replaced by a physics-space misfit

$$\sum_t \big\|\mathcal{M}^t(\boldsymbol{x}_0) - [H_\theta^{-1}(Y)]_t\big\|^2,$$

which is much closer to convex because it removes H from inside the objective. They further propose an optional *hybrid* schedule: run physics-space optimization first as a warm start, then refine against the true observations in observation space, thereby keeping the smoother physics-space basin of attraction while finishing on the genuine measurement misfit. The approach is demonstrated on two systems — Lorenz-96 (one-dimensional) and 2D Kolmogorov flow (incompressible Navier–Stokes with Kolmogorov forcing) — and the reference implementation is in JAX/Flax with a JAX-CFD solver for the flow.

Crucially for our work, the original study makes two simplifications. It assimilates *clean* observations with no added measurement noise, and it minimizes an objective that contains *only* the observation-misfit term: there is no background (prior) term and no explicit observation-error covariance.

2.2 Our Extensions

Our capstone asks whether the learned-inverse-observation idea survives a more operationally realistic setting. We make three extensions, each stated precisely below; all are carried out in a from-scratch PyTorch reimplementation of the entire pipeline (the original is JAX/Flax), including the dynamical-system integrators, the CNN inverter, the correlation preconditioning, and the L-BFGS-driven DA loop.

1. **Full 4D-Var cost.** We replace the observation-misfit-only objective with the complete 4D-Var cost $J(\boldsymbol{x}_0) = J_\mathrm{b} + J_\mathrm{o}$ (Eq. (3)), which adds a background term and an explicit observation-error covariance, with background covariance $\boldsymbol{B} = \sigma_\mathrm{b}^2 \boldsymbol{C}$ ($\boldsymbol{C}$ the system's spatial correlation) and observation-error covariance $\boldsymbol{R} = \sigma_\mathrm{obs}^2 \boldsymbol{I}$. The background term J_b pulls the analysis toward a prior $\boldsymbol{x}_b$ and makes the problem well-posed when observations are sparse; σ_b is the background-error standard deviation we tune and σ_obs is the observation-error standard deviation. The physics-space hybrid companion cost analogously replaces J_o with $\frac{1}{2\sigma_p^2} \sum_t \|\mathcal{M}^t(\boldsymbol{x}_0) - [H_\theta^{-1}(Y)]_t\|^2$.

2. **Observation noise.** Every experiment now corrupts the observations with independent additive Gaussian noise, $\boldsymbol{y} = H(\boldsymbol{x}) + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma_\mathrm{obs}^2 \boldsymbol{I})$, at noise levels $\sigma_\mathrm{obs} \in \{0.1, 0.5, 1.0\}$ (the original paper used clean observations). This tests robustness and also probes a deliberate mismatch, since the inverter is trained at one noise level and evaluated at others.

3. **Sliding-window (cycling) 4D-Var.** We add an operational cycling extension in which a short assimilation window is advanced through time and each cycle's analysis is propagated forward to form the background for the next window — the way 4D-Var is actually run operationally — in contrast to the paper's single fixed window.

Two implementation choices that the Methods section develops are worth flagging here so the cost above reads unambiguously. For Lorenz-96 the optimization is carried out in decorrelated coordinates $\boldsymbol{z} = \boldsymbol{C}^{-1/2}\boldsymbol{x}$, which turns the background term into the isotropic $\|\boldsymbol{z}_0 - \boldsymbol{z}_b\|^2/(2\sigma_\mathrm{b}^2)$; the measured spatial-correlation matrix $\boldsymbol{C}$ has condition number 8.72, and the states have climatological mean near 2.34. For 2D Kolmogorov flow we optimize the vorticity field directly with $\boldsymbol{B} = \sigma_\mathrm{b}^2 \boldsymbol{I}$ (no decorrelation), because forming and factorizing a full 4096×4096 spatial covariance is prohibitively expensive. We report results honestly, separating "reproduced the paper" from "our extensions" and flagging negative findings and smoke-scale caveats (small ensembles, the CPU-limited Kolmogorov runs, the noise-level mismatch above, and a short Lyapunov time induced by the fixed-step integrator) where they arise.

3 Models and Methods

This section describes the two dynamical systems we assimilate, the full 4D-Var cost we minimize, the learned inverse observation operator and its two convolutional variants, our observation-noise model, the correlation preconditioning, the optimizer, and the sliding-window cycling scheme. Throughout, $\mathcal{M}^t(\boldsymbol{x}_0)$ denotes the model state at outer step t integrated from the initial condition $\boldsymbol{x}_0$, H is the (sparse, subsampling) observation operator, and $\boldsymbol{y}_t = H\mathcal{M}^t(\boldsymbol{x}_0)+\boldsymbol{\varepsilon}_t$ the corresponding noisy observation.

3.1 Dynamical systems

The first testbed is the 1D Lorenz-96 model, a periodic chain of $K = 40$ grid points evolving under the ordinary differential equation (ODE)

$$\dot{x}_k = (x_{k+1} - x_{k-2})\, x_{k-1} - x_k + F, \qquad x_{k+K} = x_k, \tag{1}$$

with forcing $F = 8$ and cyclic indices. We integrate Eq. (1) with a hand-written fixed-step fourth-order Runge–Kutta (RK4) scheme in which each outer step $\Delta t_{\text{out}} = 0.1$ MTU is composed of ten inner RK4 substeps of size $\Delta t_{\text{in}} = 0.01$. The observation operator H subsamples every fourth grid point, so the 40-dimensional state is observed at only 10 sites.

To check that this fixed-step integrator is faithful at the short horizons relevant to assimilation, we compared it against an adaptive Dormand–Prince (dopri5) solver. Starting from identical attractor states, the L^2 trajectory divergence stays near machine precision while $t \ll T_L$ and first reaches the order-one (state-norm) scale at a median $t^\star = 7.70$ MTU (range $[6.50, 8.30]$ over five warmed-up states; 6.30 MTU for a single trajectory), saturating near 28.5 once the trajectories fully decorrelate by $t \approx 19.9$ MTU. A direct two-trajectory separation fit (initial separation 10^{-6}, fitted over $t \in [0.5, 5.0]$ MTU) gives a leading exponent $\lambda = 2.263\,\text{MTU}^{-1}$ and a Lyapunov time $T_L = 1/\lambda = 0.442$ MTU. This is shorter than the value of ~ 0.6 MTU expected from the literature, a known consequence of the coarse fixed-step integrator that we report honestly; our single-window experiments use a window of $T = 10$ outer steps (1.0 MTU), comfortably inside $t^\star$, so this integrator choice does not contaminate the assimilation results.

The second testbed is 2D Kolmogorov flow: the vorticity form of the incompressible Navier–Stokes equations (a partial differential equation, PDE) on the periodic domain $[0, 2\pi]^2$ with sinusoidal Kolmogorov forcing at wavenumber $k = 4$, viscosity $\nu = 10^{-2}$, and linear (Rayleigh) drag $\alpha = 0.1$. The state is a single vorticity field ω on a 64×64 grid (4096 degrees of freedom), advanced with a pseudo-spectral RK4 solver at outer step 0.18 composed of 25 inner substeps. Here H subsamples every sixteenth point in each direction, giving a 4×4 observation grid (16 of 4096 points observed). Warmed-up initial conditions are produced by integrating a spectrally filtered random field forward 100 outer steps; a warmup diagnostic confirms the flow reaches an approximately quasi-stationary turbulent regime, with the enstrophy $\frac{1}{2}\langle\omega^2\rangle$ rising from 0.50 at the smooth cold start to ≈ 9.8 near the warmup horizon (last-20-step relative drift $\approx 10.8\%$, i.e. approximately quasi-stationary) and the training-trajectory vorticity spanning roughly $[-12.3, 14.1]$.

3.2 Observation-noise model

Departing from the original study, which assimilated clean observations, every experiment here corrupts the observations with independent additive Gaussian noise,

$$\boldsymbol{y} = H(\boldsymbol{x}) + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}\big(\mathbf{0},\, \sigma_{\text{obs}}^2 \boldsymbol{I}\big), \tag{2}$$

applied independently at each observed site and time, so that the observation-error covariance is $\boldsymbol{R} = \sigma_{\mathrm{obs}}^2 \boldsymbol{I}$. For the Lorenz-96 single-window experiments we use $\sigma_{\mathrm{obs}} \in \{0.1, 0.5, 1.0\}$, spanning a range from a sharp, nearly-clean perturbation to a level comparable to the Lorenz-96 state amplitude; Figure 1 plots the three corresponding probability density functions. The sliding-window runs use $\sigma_{\mathrm{obs}} \in \{0.0, 0.5\}$ for Lorenz-96 and $\{0.0\}$ for Kolmogorov flow. In the cost, σ_{obs} doubles as the inverse-variance weight on the observation term; the perfect-observation case is handled by flooring it (at 0.1 for Lorenz-96 and 0.05 for Kolmogorov flow) so that J_{o} remains finite.

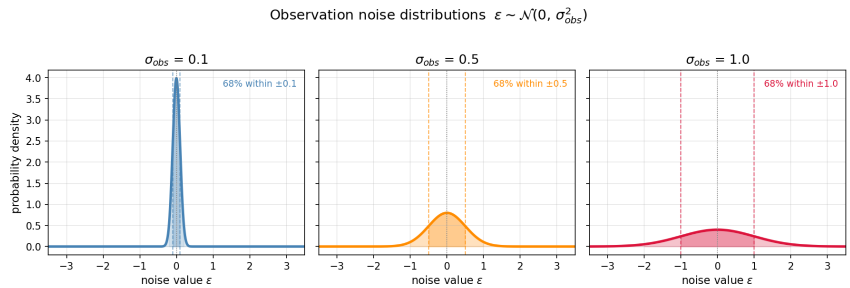

Figure 1: Gaussian observation-noise probability density functions used to corrupt the observations in every experiment. Each curve is the density of $\varepsilon \sim \mathcal{N}(0, \sigma_{\mathrm{obs}}^2)$ for one of the three noise standard deviations $\sigma_{\mathrm{obs}} \in \{0.1, 0.5, 1.0\}$; the horizontal axis is the noise value added to a single observed component and the vertical axis is the probability density. Each observed site and time receives an independent draw, so the observation-error covariance is $\boldsymbol{R} = \sigma_{\mathrm{obs}}^2 \boldsymbol{I}$. The $\sigma_{\mathrm{obs}} = 0.1$ case is a sharp, nearly-clean perturbation, whereas $\sigma_{\mathrm{obs}} = 1.0$ broadens to a level comparable to the Lorenz-96 state amplitude, making observations substantially less informative and stress-testing the assimilation methods.

3.3 The full 4D-Var cost

Our central extension replaces the paper's bare observation-misfit objective with the complete 4D-Var cost, $J(\boldsymbol{x}_0) = J_{\mathrm{b}} + J_{\mathrm{o}}$, where

$$J(\boldsymbol{x}_0) = \underbrace{\tfrac{1}{2}(\boldsymbol{x}_0 - \boldsymbol{x}_b)^\top \boldsymbol{B}^{-1}(\boldsymbol{x}_0 - \boldsymbol{x}_b)}_{J_{\mathrm{b}}} + \underbrace{\tfrac{1}{2}\sum_t (H\mathcal{M}^t(\boldsymbol{x}_0) - \boldsymbol{y}_t)^\top \boldsymbol{R}^{-1}(H\mathcal{M}^t(\boldsymbol{x}_0) - \boldsymbol{y}_t)}_{J_{\mathrm{o}}}, \tag{3}$$

with background covariance $\boldsymbol{B} = \sigma_{\mathrm{b}}^2 \boldsymbol{C}$ ($\boldsymbol{C}$ the spatial correlation), observation-error covariance $\boldsymbol{R} = \sigma_{\mathrm{obs}}^2 \boldsymbol{I}$, and $\boldsymbol{x}_b$ the background (prior) estimate. The background term J_{b} pulls the analysis toward $\boldsymbol{x}_b$ and is what makes the problem well-posed when observations are sparse (10 of 40 for Lorenz-96, 16 of 4096 for Kolmogorov flow); σ_{b} is the background-error standard deviation we tune.

Decorrelation preconditioning (Lorenz-96 only). We estimate $\boldsymbol{C}$ as the spatial covariance of an ensemble of warmed-up states (condition number 8.72, climatological mean ≈ 2.34) and optimize in decorrelated coordinates $\boldsymbol{z} = \boldsymbol{C}^{-1/2}\boldsymbol{x}$, obtained from a symmetric eigendecomposition. With $\boldsymbol{B} = \sigma_{\mathrm{b}}^2 \boldsymbol{C}$ this reduces the background term to the isotropic $J_{\mathrm{b}} = \|\boldsymbol{z}_0 - \boldsymbol{z}_b\|^2/(2\sigma_{\mathrm{b}}^2)$; the starting point is decorrelated before optimization and the optimized $\boldsymbol{z}_0$ is re-correlated to physics coordinates at the end. For Kolmogorov flow we instead optimize the vorticity field directly with an identity background, $\boldsymbol{B} = \sigma_{\mathrm{b}}^2 \boldsymbol{I}$ and $J_{\mathrm{b}} = \|\omega_0 - \omega_b\|^2/(2\sigma_{\mathrm{b}}^2)$, because forming and factorizing a full 4096×4096 spatial covariance is prohibitively expensive.

3.4 Learned inverse observation operator

Following the paper's key idea, we train a CNN H_θ^{-1} that acts as a learned inverse observation operator, mapping a stacked (time, space) tensor formed from an observation sequence $Y \in \mathbb{R}^{T \times n_{\text{obs}}}$ back to a full-state sequence $X \in \mathbb{R}^{T \times n_{\text{grid}}}$. Its building block is a periodic-space convolution: the spatial axis receives *circular* padding (matching the periodic boundary conditions) while the time axis receives *zero* padding, after which the low-resolution observation grid is upsampled to the full grid.

We implement two PyTorch variants. The *original-port* inverter uses a hidden width of 32 over six residual periodic-convolution blocks with Gaussian error linear unit (GELU) activations and bilinear spatial upsampling. The *paper-faithful* inverter follows the architecture of Frerix et al. [1] more closely: sigmoid linear unit (SiLU) activations, batch normalization (BN) after every convolution, cubic periodic upsampling, and a decreasing channel pyramid of widths 128/64/32/16 with no skip connections. Both are trained by supervised regression — integrate trajectories, apply H (plus noise), and fit the network to recover the full state under a mean-squared-error (MSE) loss with the Adam optimizer. Trained at $\sigma_{\text{obs}} = 0$ on 32,000 Lorenz-96 trajectories, the two variants reach comparable reconstruction quality (held-out test L_1 ratio 1.165), and the paper-faithful inverter is retrained at each noise level to supply a noise-matched lift for the hybrid runs.

Physics-space hybrid companion cost. The learned inverse enables a second, more nearly convex objective that replaces the lossy observation misfit J_{o} with a full-state misfit against the lifted observations,

$$J_{\text{phys}}(\boldsymbol{x}_0) = J_{\text{b}} + \frac{1}{2\sigma_p^2} \sum_t \left\| \mathcal{M}^t(\boldsymbol{x}_0) - [H_\theta^{-1}(Y)]_t \right\|^2, \tag{4}$$

where $H_\theta^{-1}(Y)$ lifts the observation sequence into physics space, removing the many-to-one operator H from inside the objective, and σ_p is the physics-space error scale (set to $\sigma_p = 0.1$ in our runs). The *hybrid* schedule runs physics-space optimization first as a warm start (Eq. (4)) and then refines against the true observations with the observation-space cost $J_{\text{b}} + J_{\text{o}}$ (Eq. (3)); we contrast it throughout against pure observation-space optimization. The learned operator is also used as an initializer: the cold-start guess $H_\theta^{-1}(Y)$ replaces a naive interpolation or climatological-mean baseline.

3.5 Optimization

All variational sub-problems are solved with the L-BFGS quasi-Newton method using a strong-Wolfe line search. The entire N-sample ensemble is optimized in a single batched L-BFGS call: because each row's gradient depends only on its own observations, the batched problem is exactly equivalent to N independent 4D-Var problems. The hybrid schedule allocates a fixed number of physics-space iterations followed by observation-space iterations (for the single-window Lorenz-96 runs, 200 then 300; for Lorenz-96 cycling, 100 then 400), whereas pure observation-space runs spend the full iteration budget on $J_{\text{b}} + J_{\text{o}}$.

3.6 Sliding-window (cycling) 4D-Var

We extend the single-window setup to an operational cycling scheme. A fixed-length assimilation window (Lorenz-96 `WINDOW_T`= 8; Kolmogorov `WINDOW_T`= 10 outer steps) slides over a long observation record with a configurable *stride* controlling window overlap (Lorenz-96 strides $\{2, 4, 8\}$

for $T_{\text{obs}} \in \{24, 48\}$; Kolmogorov strides $\{2, 5, 10\}$ for $T_{\text{obs}} \in \{20, 30\}$). The two roles of the background are decoupled: on cycle 0 the J_{b} background equals the cold-start initializer, while for cycles ≥ 1 the background is the *propagated previous analysis* — the prior cycle's optimized state integrated forward to the new window start. Each cycle re-initializes the L-BFGS starting point from a fresh H_θ^{-1} estimate of the current window, and the noise-matched inverter checkpoint is used at each cycle. The stride is chosen so that the most recent window is always assimilated even when ($T_{\text{obs}} - \texttt{WINDOW\_T}$) is not divisible by it. Because cycling trusts the propagated analysis, it uses a tighter background-error standard deviation ($\sigma_{\text{b}} = 0.3$) than the cold-start single window ($\sigma_{\text{b}} = 1.0$).

4 Results

We present results for the two test systems in turn: first the one-dimensional Lorenz-96 model, then two-dimensional Kolmogorov flow.

4.1 Lorenz-96

We organize the Lorenz-96 results as follows: first the integrator and Lyapunov diagnostics that establish the chaotic time scale of the testbed; then the training of the learned inverse observation operator H_θ^{-1}; then the choice of cold-start initialization; then our central single-window result, the tuned-σ_{b} full 4D-Var sweep; and finally the sliding-window (cycling) experiments.

4.1.1 Integrator and Lyapunov diagnostics

Our PyTorch port integrates Lorenz-96 (grid size 40, forcing $F = 8$) with a fixed-step RK4 scheme, ten inner substeps of $\Delta t_{\text{in}} = 0.01$ per outer step $\Delta t_{\text{out}} = 0.1$, whereas the reference paper uses an adaptive dopri5 solver. To check that this choice does not contaminate the DA results, we propagate the same warmed-up attractor state 200 outer steps (20 MTU) with both integrators and measure the L^2 state divergence $\|\boldsymbol{x}_{\text{RK4}}(t) - \boldsymbol{x}_{\text{adapt}}(t)\|_2$ (Figure 2). Because the global RK4 truncation error is $O(\Delta t_{\text{in}}^4) = O(10^{-8})$, well below the adaptive tolerances, the two integrators agree to near machine precision while $t \ll T_L$ and only diverge once the tiny per-step difference is amplified at the chaotic rate. The divergence first reaches the order-one (state-norm) scale at $t^* = 6.30$ MTU for a single trajectory; across five warmed-up states the median is $t^* = 7.70$ MTU (range $[6.50, 8.30]$), and the divergence saturates near 28.5 by $t = 19.9$ MTU once the trajectories fully decorrelate. A direct two-trajectory separation fit (initial separation $\epsilon_0 = 10^{-6}$, fit window $t \in [0.5, 5.0]$ MTU; Figure 3) gives a leading growth rate $\lambda = 2.263\,\text{MTU}^{-1}$ and Lyapunov time $T_L = 1/\lambda = 0.442$ MTU. This is somewhat shorter than the paper's expected ~ 0.6 MTU; we report it honestly as a known consequence of the fixed-step integrator being slightly too coarse to resolve the fastest unstable directions. The practical upshot is that t^* is the operational ceiling on assimilation-window length over which the two simulators represent the same physics: our single-window experiments use $T = 10$ outer steps ($= 1.0$ MTU), comfortably inside t^*, so the integrator choice does not affect the DA conclusions below.

4.1.2 Inverse-operator training

The learned inverse observation operator H_θ^{-1} maps the observation sequence Y (only 10 of 40 grid points, since H subsamples every fourth point) back to the full physical-state sequence X. We train two CNN variants on the same 32,000-trajectory dataset with identical Adam settings (500 epochs,

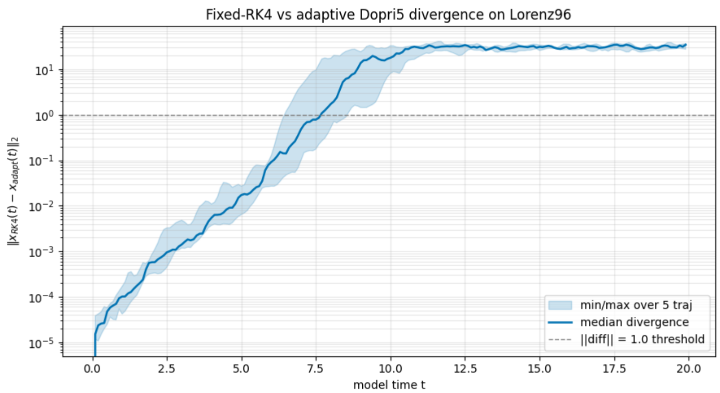

Figure 2: Fixed-step RK4 versus adaptive dopri5 integrator divergence on Lorenz-96. Vertical axis (log scale): the L^2 state difference $\|\boldsymbol{x}_{\mathrm{RK4}}(t) - \boldsymbol{x}_{\mathrm{adapt}}(t)\|_2$; horizontal axis: model time t in MTU. The solid blue curve is the median over five warmed-up attractor states, the shaded band the min/max envelope, and the dashed grey line marks the order-one threshold. The two integrators agree to near machine precision while $t \ll T_L$, then diverge at the chaotic rate, crossing the order-one scale at a median $t^* = 7.70$ MTU and saturating near 28.5. Our single-window experiments use $T = 1.0$ MTU, comfortably inside t^*, so the integrator choice does not affect the DA results.

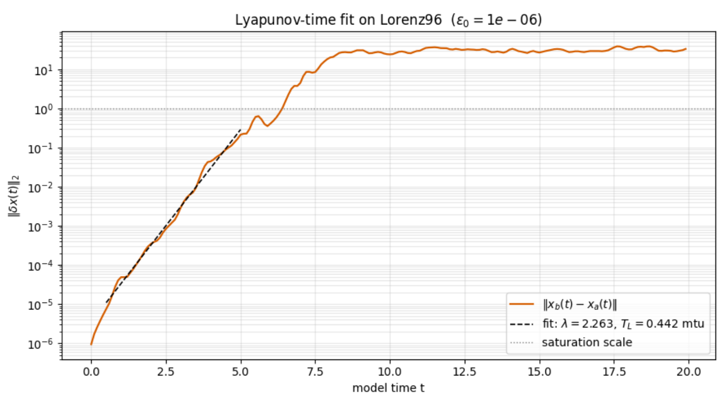

Figure 3: Direct Lyapunov-time estimate for Lorenz-96. Two trajectories are launched $\epsilon_0 = 10^{-6}$ apart and the L^2 separation $\|\delta\boldsymbol{x}(t)\|_2$ (orange, log scale) is plotted against model time t; the dashed black line is an exponential fit over the linear-growth window $t \in [0.5, 5.0]$ MTU and the dotted grey line marks the saturation scale. The fit yields growth rate $\lambda = 2.263\,\mathrm{MTU}^{-1}$ and Lyapunov time $T_L = 1/\lambda = 0.442$ MTU, after which the separation saturates near the attractor diameter. This is somewhat shorter than the paper's expected $\sim$0.6 MTU, a known consequence of the fixed-step RK4 integrator being slightly too coarse to resolve the fastest unstable directions.

batch size 256, learning rate 10^{-3}). The original PyTorch port (GELU activations; flat width-32 residual stack; bilinear upsampling) reaches a final training MSE of 0.3714, while the paper-faithful inverter (SiLU activations; channel pyramid 128→64→32→16; BN after every convolution; cubic-periodic upsampling; no skip connections) reaches 0.4924 (Figure 4). On a held-out 100-trajectory set (seed 100), the port attains test $L_1 = 0.3838$ with $\boldsymbol{x}_0$ RMSE 0.9973 ± 0.3261, and the paper-faithful inverter attains test $L_1 = 0.4470$ with $\boldsymbol{x}_0$ RMSE 1.1635 ± 0.4036. The paper/port L_1 ratio of 1.165 lies comfortably within a 30% acceptance band, so the paper-faithful architecture reproduces the port's reconstruction quality despite the architectural differences. Retraining the paper-faithful inverter at higher noise gives final MSE 0.6321, 1.3338, and 2.2201 at $\sigma_{\mathrm{obs}} = 0.1, 0.5, 1.0$ respectively; these noise-matched checkpoints supply the inverter used in the hybrid DA runs below.

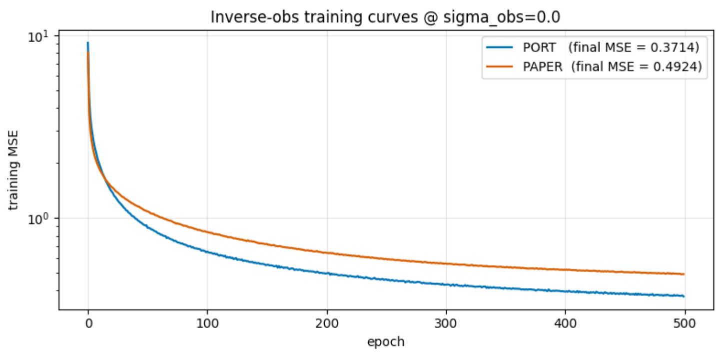

Figure 4: Training-loss curves for the two inverse observation operators at $\sigma_{\mathrm{obs}} = 0$. Both networks are trained on the same 32,000-trajectory dataset with identical Adam settings (500 epochs); vertical axis: training MSE (log scale); horizontal axis: epoch. The blue curve is the original PyTorch port (GELU activations, flat width-32 residual stack, bilinear upsampling), converging to final MSE 0.3714; the orange curve is the paper-faithful inverter (SiLU activations, $128 \to 64 \to 32 \to 16$ channels, BN after every convolution, cubic-periodic upsampling, no skip connections), converging to 0.4924. The two curves track closely, and despite the architectural differences the paper-faithful inverter reproduces the port's reconstruction quality (held-out test L_1 ratio 1.165, within the 30% acceptance band).

4.1.3 Baseline initialization

A 4D-Var solve needs a cold-start initial guess for the optimizer. We compare the original "repeat-interleave" baseline (copy each $t = 0$ observation across its block of four unobserved points, producing step discontinuities at observation boundaries) against the paper's climatological-mean baseline (fill unobserved positions with the attractor mean $\bar{X} \approx 2.34$, then overwrite observed positions with $\boldsymbol{y}_0$). At $\sigma_{\mathrm{obs}} = 0$ ($T = 10$, $N = 100$), the raw initializers give direct $\boldsymbol{x}_0$ RMSE 4.7828 ± 0.5006 (repeat) versus 3.0892 ± 0.2539 (climatological mean), a ratio of 0.646, i.e. the climatological-mean prior is roughly 35% closer to truth. The spatial-correlation matrix $\boldsymbol{C}$ used for decorrelation preconditioning has condition number 8.72. Figure 5 shows analysis RMSE for the four (initializer $\times$ optimization-mode) combinations under untuned full 4D-Var ($\sigma_{\mathrm{b}} = 1.0$ fixed): both a better cold-start prior and hybrid optimization lower the analysis error before σ_{b} is even tuned, with the climatological-mean baseline plus hybrid optimization lowest at every noise level. The space-time (Hovmöller) comparison at $\sigma_{\mathrm{obs}} = 0.5$ (Figure 6) makes the qualitative difference concrete: observation-space 4D-Var from the repeat-interleave baseline carries visible artifacts and a distorted wave pattern out of the assimilation window, whereas the climatological-mean baseline reproduces the truth's traveling-wave structure far more faithfully into the forecast.

4.1.4 Tuned-σ_{b} full 4D-Var: the key result

Our central single-window experiment sweeps the background-error standard deviation σ_{b} over $\{0.1, 0.3, 1.0, 3.0, 10.0\}$ for every (initializer $\times$ optimization-mode $\times$ noise) cell of the full 4D-Var cost $J = J_{\mathrm{b}} + J_{\mathrm{o}}$, with $\boldsymbol{B} = \sigma_{\mathrm{b}}^2 \boldsymbol{C}$ and $\boldsymbol{R} = \sigma_{\mathrm{obs}}^2 \boldsymbol{I}$, and records the lowest-RMSE choice ($N = 16$,

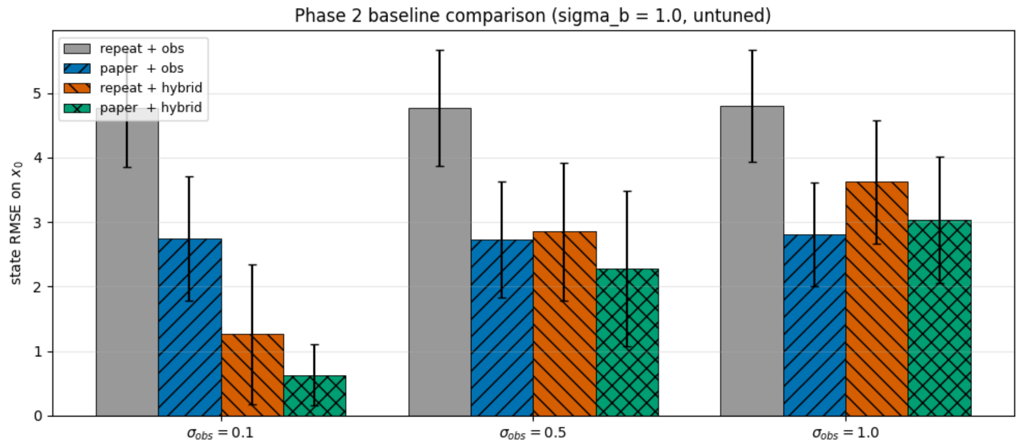

Figure 5: Analysis RMSE comparison of the two baseline initializations under untuned full 4D-Var ($\sigma_\mathrm{b} = 1.0$ fixed). Grouped bars show post-assimilation state RMSE on $\boldsymbol{x}_0$ ($\pm$ one standard deviation over $N = 100$ trajectories) for the four combinations of initializer (repeat-interleave vs. climatological-mean) and optimization mode (observation-space vs. hybrid), across noise levels $\sigma_\mathrm{obs} \in \{0.1, 0.5, 1.0\}$. The repeat-interleave baseline with observation-only optimization (grey) is worst at every noise level; the climatological-mean baseline with hybrid optimization is lowest (most clearly at $\sigma_\mathrm{obs} = 0.1$). A better cold-start prior and hybrid optimization each reduce analysis error even before σ_b is tuned, motivating the per-cell σ_b sweep that follows.

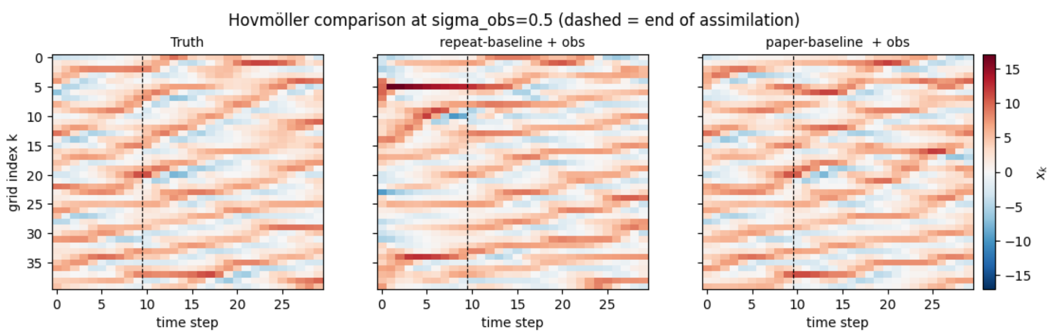

Figure 6: Hovmöller comparison of the two baselines at $\sigma_\mathrm{obs} = 0.5$. Each panel shows the Lorenz-96 state x_k (color, red–blue diverging scale) with grid index k on the vertical axis and time step on the horizontal axis: ground-truth trajectory (left), observation-space 4D-Var initialized from the repeat-interleave baseline (center), and from the climatological-mean baseline (right). The dashed vertical line marks the end of the 10-step assimilation window; everything to its right is free forecast. The repeat baseline carries visible artifacts and a distorted wave pattern out of the window, whereas the climatological-mean baseline reproduces the truth's traveling-wave structure more faithfully into the forecast under moderate observation noise.

$T = 10$). Table 1 reports the best analysis RMSE per cell with the selected σ_b in parentheses, and Figure 7 plots the same data with the optimal σ_b annotated.

Table 1: Best analysis RMSE on $\boldsymbol{x}_0$ for tuned-σ_{b} full 4D-Var on Lorenz-96 ($N = 16$, $T = 10$), with the selected $\sigma_{\text{b}} \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$ in parentheses. "paper" denotes the climatological-mean baseline and "invobs" the learned-inverse initialization; "obs" is observation-space optimization and "hybrid" the physics-space-then-observation-space schedule. Bold marks the lowest RMSE in each noise column.

Method	$\sigma_{\text{obs}} = 0.1$	$\sigma_{\text{obs}} = 0.5$	$\sigma_{\text{obs}} = 1.0$
paper + obs	2.466 (0.1)	2.393 (0.3)	2.708 (0.3)
paper + hybrid	0.621 (3.0)	1.223 (1.0)	1.846 (1.0)
invobs + obs	0.883 (0.3)	1.550 (0.3)	1.867 (1.0)
invobs + hybrid	**0.620** (10.0)	**1.168** (1.0)	**1.676** (1.0)

Three findings stand out. First, the positive headline: invobs initialization combined with hybrid (physics-space then observation-space) optimization attains the lowest analysis RMSE at every noise level (0.620, 1.168, 1.676 for $\sigma_{\text{obs}} = 0.1, 0.5, 1.0$), confirming that the paper's central idea survives in our full 4D-Var setting with observation noise (at $\sigma_{\text{obs}} = 0.1$ this is effectively a tie with paper+hybrid, 0.620 vs 0.621). Second, hybrid optimization dramatically beats observation-only optimization for the climatological-mean baseline that has no learned lift: at $\sigma_{\text{obs}} = 0.5$, paper+obs (2.393) is roughly halved to paper+hybrid (1.223), because observation-only 4D-Var from a poor prior falls into bad local minima when only 10 of 40 points are observed, and the physics-space warm start escapes them. Third, an honest negative result: the expected ordering—that invobs+hybrid should prefer a *smaller* optimal σ_{b} than paper+obs—does not hold. invobs+hybrid selects $\sigma_{\text{b}} = 10.0/1.0/1.0$ while paper+obs selects $\sigma_{\text{b}} = 0.1/0.3/0.3$, the opposite direction, failing the acceptance check at all three noise levels. We also note a secondary caveat: tuning σ_{b} helped only marginally over the untuned $\sigma_{\text{b}} = 1$ in most cells (gains typically below 0.2 RMSE, and exactly zero where the best σ_{b} was already 1.0), and the small ensemble ($N = 16$) makes the error bars wide.

4.1.5 Sliding-window (cycling) 4D-Var

We next extend the single fixed window to an operational cycling scheme. All Lorenz-96 sliding-window experiments use grid size 40, H observing every fourth point, per-cycle window `WINDOW_T` $=$ 8 steps, a 50-step forecast horizon evaluated after the last window, and an ensemble of $N = 50$ trajectories at noise levels $\sigma_{\text{obs}} \in \{0.0, 0.5\}$. Single-window (cold-start) runs use background-error standard deviation $\sigma_{\text{b}} = 1.0$, while cycling runs use a tighter $\sigma_{\text{b}} = 0.3$ because the cycled background—the previous analysis propagated forward to the new window start—is trusted more than a cold start. Optimization is L-BFGS in decorrelated coordinates $\boldsymbol{z} = \boldsymbol{C}^{-1/2}\boldsymbol{x}$ ($\boldsymbol{C}$ condition number 8.72); hybrid mode runs 100 physics-space iterations then 400 observation-space iterations. The roles of "L-BFGS starting point" and "J_{b} background" are decoupled: every cycle restarts the optimizer from a fresh H_θ^{-1} estimate of the new window, while for cycles ≥ 1 the J_{b} background is the propagated previous analysis (cycle 0 is a cold start whose background equals its starting point).

Experiment A (single 8-step window). Figure 8 assimilates one 8-step window at $\sigma_{\text{b}} = 1.0$ for the four (initialization $\times$ optimization) combinations. All curves drop to low RMSE inside the window and then grow toward the Lorenz-96 climatological saturation (RMSE ≈ 5) over the chaotic forecast, consistent with the short Lyapunov time ($T_L \approx 0.44$ MTU). The analysis-RMSE bars at $t = 0$ are the clean discriminator: the three optimization-capable methods reach analysis

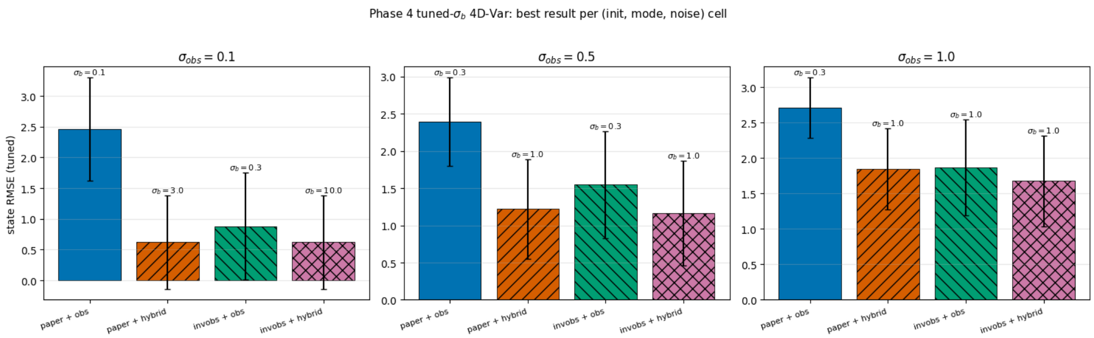

Figure 7: KEY RESULT—tuned-σ_b full 4D-Var on Lorenz-96. One panel per observation-noise level ($\sigma_\mathrm{obs} = 0.1, 0.5, 1.0$, left to right); within each panel the four bars are the (initializer $\times$ optimization-mode) combinations paper+obs, paper+hybrid, invobs+obs, invobs+hybrid, showing the lowest analysis RMSE on $\boldsymbol{x}_0$ ($\pm$ one standard deviation, $N = 16$) after sweeping $\sigma_\mathrm{b} \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$; the optimal σ_b is annotated above each bar. The learned-inverse initialization with hybrid optimization (rightmost bar) attains the lowest RMSE at every noise level (0.620, 1.168, 1.676), and hybrid optimization dramatically improves the climatological-mean baseline over observation-only (paper+obs 2.393 $\rightarrow$ paper+hybrid 1.223 at $\sigma_\mathrm{obs} = 0.5$). The annotated optimal σ_b values also show the honest negative result discussed in the text: invobs+hybrid selects $\sigma_\mathrm{b} = 10.0/1.0/1.0$ versus paper+obs 0.1/0.3/0.3, the opposite of the expected ordering.

RMSE ≈ 1 at $\sigma_\mathrm{obs} = 0$ and ≈ 1.6–1.9 at $\sigma_\mathrm{obs} = 0.5$, while baseline+obs-only stays near 5 at both noise levels—the same observation-space local-minimum failure documented above, which hybrid (physics-space) warm-starting rescues. By lowest mean forecast RMSE the best single-window method is baseline+hybrid at $\sigma_\mathrm{obs} = 0$ and invobs+hybrid at $\sigma_\mathrm{obs} = 0.5$; these selections feed the references in the experiments below. The $\sigma_\mathrm{obs} = 0$ margin among the three good methods is within ensemble scatter, so that "best" label is a narrow tie-break.

Experiment B1 (stride sweep). For each observation history $T_\mathrm{obs} \in \{24, 48\}$ and $\sigma_\mathrm{obs} \in \{0, 0.5\}$ we run cycling with strides $\{2, 4, 8\}$ (cycle counts $\{9, 5, 3\}$ for $T_\mathrm{obs} = 24$ and $\{21, 11, 6\}$ for $T_\mathrm{obs} = 48$), all with invobs init each cycle, hybrid optimization, propagated background, and $\sigma_\mathrm{b} = 0.3$. Figure 9 shows the three stride curves are nearly indistinguishable within ensemble scatter and all relax to the same forecast saturation; the differences are confined to the window and shortest lead times. By mean forecast RMSE the selected best strides are 8, 2, 8, 8 for $(T_\mathrm{obs}, \sigma_\mathrm{obs}) = (24, 0), (24, 0.5), (48, 0), (48, 0.5)$—narrow tie-breaks among overlapping curves rather than strong preferences. The dashed single-window reference sits on top of the cycling curves over most of the forecast, foreshadowing that cycling's advantage is modest and concentrated near the analysis time.

Experiment B2 (per-cycle error reduction). For each panel's best-stride run, Figure 10 plots window-start RMSE per cycle for the J_b background (the propagated previous analysis) and the analysis, with the single-window invobs+hybrid analysis RMSE as a horizontal reference. The intended mechanism is visible: the analysis sits at or below the background within each cycle (the assimilation reduces error), the background error generally falls as cycling accumulates history, and after the first cycle the cycled analysis drops below the single-window reference. The descent is cleanest at $\sigma_\mathrm{obs} = 0.5$ (most clearly $T_\mathrm{obs} = 48$, stride 8, where both curves fall monotonically and the analysis ends well under the reference). The $\sigma_\mathrm{obs} = 0$ panels are noisier and non-monotone (e.g. a mid-sequence bump for $T_\mathrm{obs} = 48$): with perfect observations a single 8-step window already

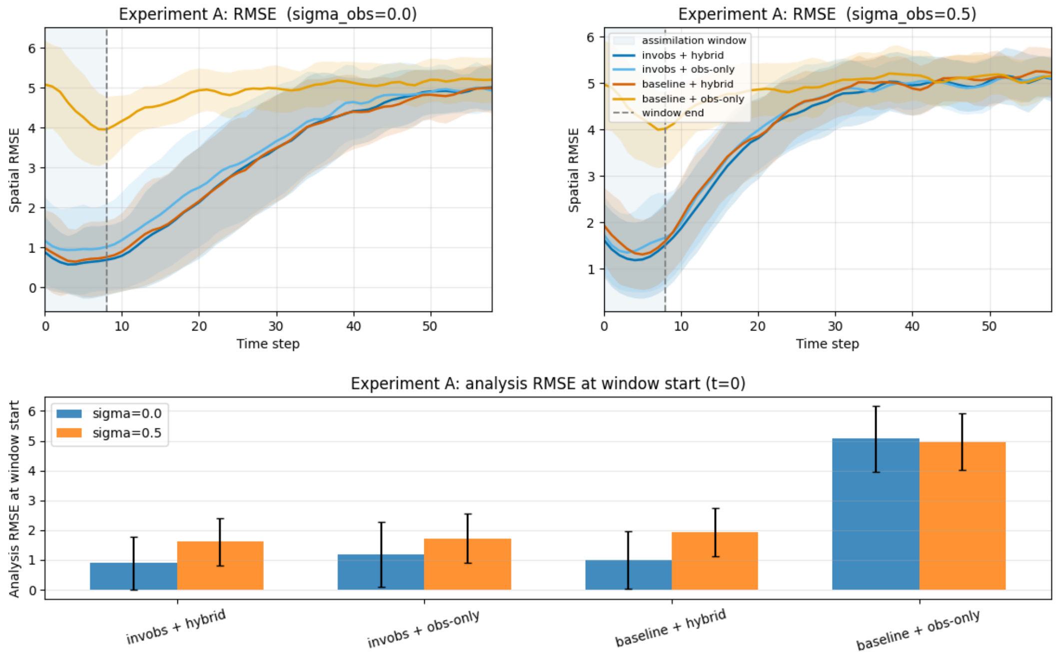

Figure 8: Experiment A: single 8-step assimilation window on Lorenz-96 ($N = 50$, $\sigma_\mathrm{b} = 1.0$), comparing four initialization $\times$ optimization combinations. **Top row:** ensemble-mean spatial RMSE versus time step for $\sigma_\mathrm{obs} = 0.0$ (left) and $\sigma_\mathrm{obs} = 0.5$ (right); the shaded band marks the assimilation window (steps 0–8), the dashed line the window end, and coloured envelopes are ± 1 ensemble standard deviation. All curves fall to low error inside the window and then grow toward the climatological saturation (RMSE ≈ 5) over the 50-step forecast, except baseline+obs-only, which never leaves the climatological level. **Bottom:** grouped bar chart of analysis RMSE at the window start ($t = 0$): the three optimization-capable methods reach ≈ 1 at $\sigma_\mathrm{obs} = 0$ and ≈ 1.6–1.9 at $\sigma_\mathrm{obs} = 0.5$, while baseline+obs-only stays near 5. Hybrid warm-starting is essential to rescue the climatological-mean baseline from a poor observation-space local minimum.

over-determines the 10 observed components, so extra cycles add little and $N = 50$ sampling noise dominates.

Experiment C (the key cycling comparison). Figure 11 is the central cycling comparison: SW-best (best-stride cycling with full observation history) versus two single-window methods that see only the last 8 observations—invobs+hybrid and baseline+obs-only—across the $(T_\mathrm{obs}, \sigma_\mathrm{obs})$ grid. baseline+obs-only is consistently the worst, stuck near the climatological RMSE through the window (the observation-space local-minimum failure again). SW-best's per-cycle analysis chain descends as history accumulates, so it enters the last-window analysis time from a lower error than the cold single-window methods. The honest, central finding is that whether cycling *clearly* wins depends on $(T_\mathrm{obs}, \sigma_\mathrm{obs})$: the advantage is most pronounced at $\sigma_\mathrm{obs} = 0.5$ with longer history $T_\mathrm{obs} = 48$ (more cycles to average down observation noise and propagate information), and marginal at $\sigma_\mathrm{obs} = 0$, where a single well-posed 8-step window already pins the observed state and the methods overlap within $N = 50$ scatter. Once into the chaotic forecast all methods converge to the same saturation regardless: cycling improves the analysis and short-lead forecast, not the Lyapunov-time-limited long-range predictability. This directly answers the experiment's question—cycling with more history beats a single long/last window mainly when observations are noisy and

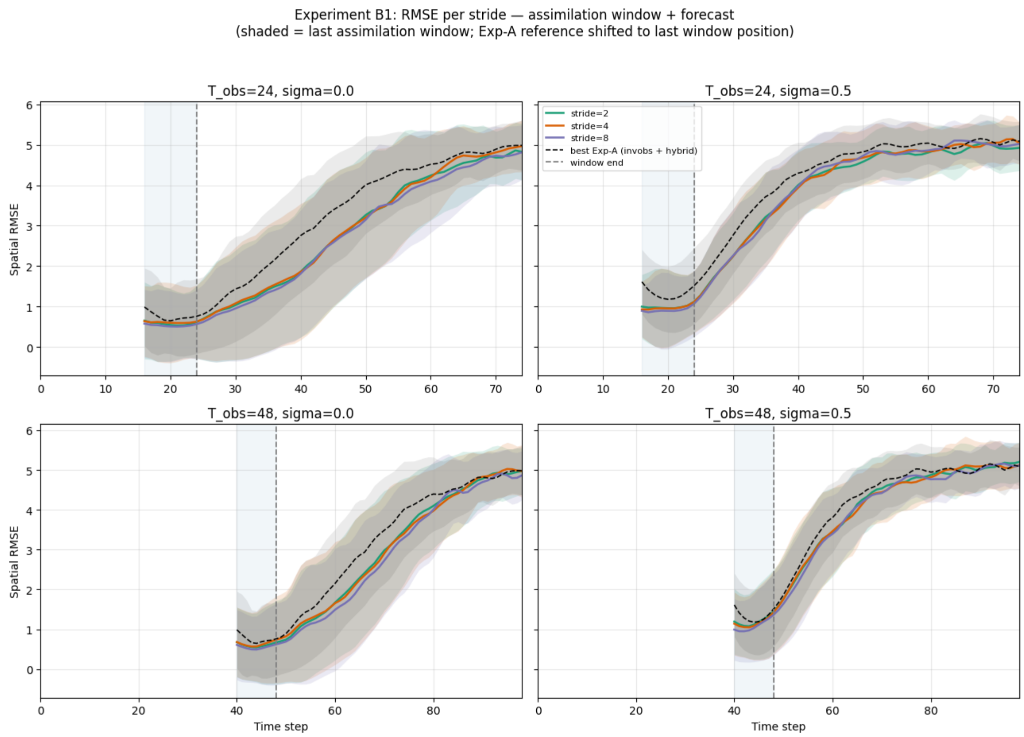

Figure 9: Experiment B1: sliding-window (cycling) 4D-Var stride sweep on Lorenz-96 ($N = 50$, invobs init each cycle, hybrid optimization, propagated-analysis background, cycling $\sigma_\mathrm{b} = 0.3$). Each panel plots last-cycle ensemble-mean spatial RMSE versus absolute time step for strides $2, 4, 8$ (coloured) with $\pm$1-std envelopes; rows are $T_\mathrm{obs} = 24$ (top) and 48 (bottom), columns are $\sigma_\mathrm{obs} = 0.0$ (left) and 0.5 (right). The shaded band is the last assimilation window $[T_\mathrm{obs}-8,\, T_\mathrm{obs}]$ and the dashed line is the window end; the dashed black curve is the best single-window Experiment-A method (baseline+hybrid at $\sigma_\mathrm{obs} = 0$, invobs+hybrid at $\sigma_\mathrm{obs} = 0.5$), time-shifted so its window end coincides with the cycling runs. The three stride curves are nearly indistinguishable within ensemble scatter and all relax to climatological saturation; by mean forecast RMSE the selected best strides are $8, 2, 8, 8$. Assimilation stride has little effect on Lorenz-96 cycling skill, so the best stride is a narrow tie-break.

history is long.

Experiment D (Hovmöller diagnostics). Finally, Figure 12 shows space-time (Hovmöller) diagnostics for one ensemble member across the four $(T_\mathrm{obs}, \sigma_\mathrm{obs})$ configurations, comparing the truth, the SW-best reconstruction and signed error, and the single-window invobs+hybrid reconstruction and signed error. Both methods reproduce the truth's traveling wave packets through and just past the window; the signed-error rows are near-white inside and immediately after the window and develop structured stripes as the chaotic forecast decorrelates. SW-best shows visibly cleaner near-window error than the cold single-window invobs+hybrid at $\sigma_\mathrm{obs} = 0.5$, illustrating the same noise-dependent cycling advantage seen in Experiment C, while error growth far into the forecast is comparable across methods. This is a single illustrative member, not a statistical summary.

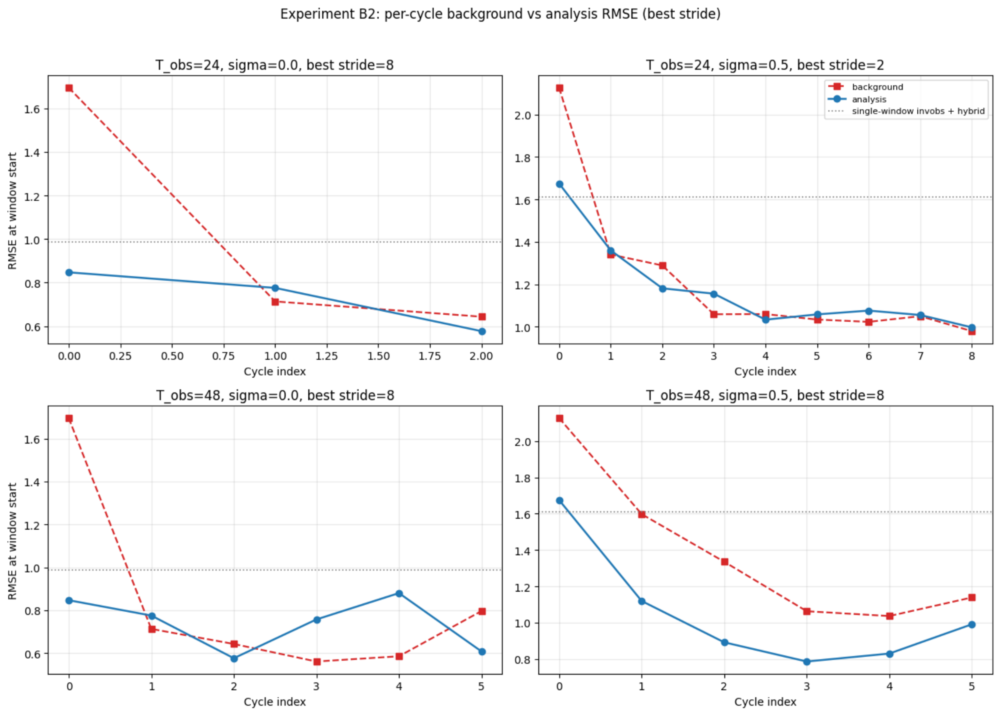

Figure 10: Experiment B2: per-cycle error reduction for the best-stride sliding-window run in each $(T_{\text{obs}}, \sigma_{\text{obs}})$ configuration ($N = 50$). Each panel plots ensemble-mean spatial RMSE at the window-start time versus cycle index for the J_{b} background (red dashed squares, the previous analysis propagated to the new window) and the analysis (blue circles), with the single-window invobs+hybrid analysis RMSE as a dotted horizontal reference. Panels: $T_{\text{obs}} = 24$ best stride 8 ($\sigma_{\text{obs}} = 0$, 3 cycles) and 2 ($\sigma_{\text{obs}} = 0.5$, 9 cycles); $T_{\text{obs}} = 48$ best stride 8 for both noise levels (6 cycles). The analysis lies at or below the background within each cycle, and after cycle 0 the cycled analysis drops below the single-window reference. The descent is cleanest at $\sigma_{\text{obs}} = 0.5$; at $\sigma_{\text{obs}} = 0$ the curves are noisier and non-monotone because a single 8-step window already over-determines the 10 observed components, leaving $N = 50$ sampling noise dominant. Cycling reduces background error into a better analysis each cycle, most clearly when observations are noisy.

4.2 Two-Dimensional Kolmogorov Flow

Our second testbed is two-dimensional (2D) Kolmogorov flow: the vorticity formulation of the incompressible Navier–Stokes equations on the periodic domain $[0, 2\pi]^2$, with sinusoidal forcing at wavenumber $k = 4$, viscosity $\nu = 10^{-2}$, and linear (Rayleigh) drag $\alpha = 0.1$. The state is a single vorticity field ω on a 64×64 grid (4096 degrees of freedom), advanced with a pseudo-spectral fourth-order Runge–Kutta (RK4) integrator at outer step 0.18. The observation operator H subsamples every sixteenth point in each direction, so only a $4 \times 4 = 16$-point subset of the 4096 grid points is observed. Unlike Lorenz-96, there is *no* correlation preconditioning here: forming and factorizing a full 4096×4096 spatial covariance is prohibitively expensive, so we optimize the vorticity ω_0 directly with an identity background, $\boldsymbol{B} = \sigma_{\text{b}}^2 \boldsymbol{I}$.

We emphasize at the outset that these are deliberately *smoke-scale* validation runs that demonstrate the pipeline mechanics, not converged scientific results. The pseudo-spectral solver relies on `torch.fft.rfft2`, which is CPU-only in our stack, so every Kolmogorov experiment runs on CPU. The sliding-window ensemble is only $N = 8$; the inverse observation operator is trained with

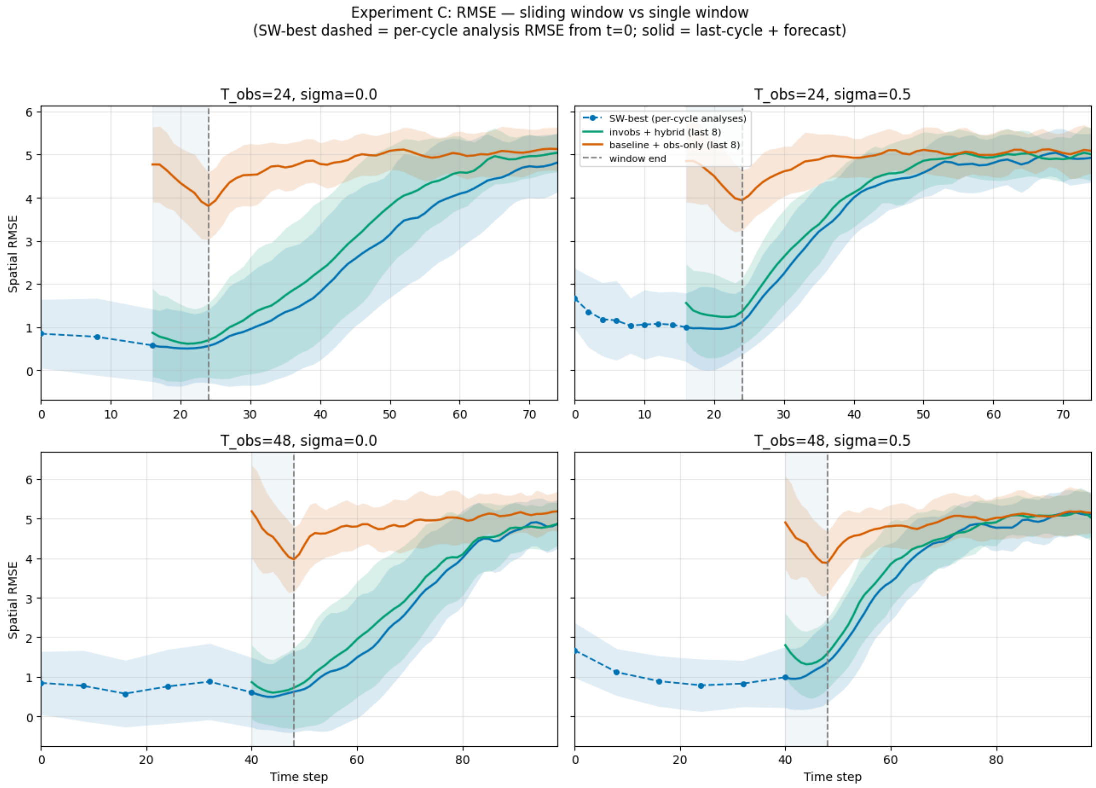

Figure 11: Experiment C (key comparison): sliding-window cycling versus single-window 4D-Var on Lorenz-96 ($N = 50$). Each $(T_{\mathrm{obs}}, \sigma_{\mathrm{obs}})$ panel (rows $T_{\mathrm{obs}} = 24, 48$; columns $\sigma_{\mathrm{obs}} = 0.0, 0.5$) plots ensemble-mean spatial RMSE versus absolute time step with ± 1-std envelopes for three methods: SW-best (best-stride cycling with full observation history), a dashed per-cycle analysis-RMSE chain from $t = 0$ that hands off to a solid continuous curve for the last cycle plus 50-step forecast; invobs+hybrid on the last 8 observations; and baseline+obs-only on the last 8 observations. The shaded band is the last assimilation window $[T_{\mathrm{obs}} - 8, T_{\mathrm{obs}}]$ and the dashed line marks the window end. baseline+obs-only stays near the climatological RMSE throughout the window; SW-best's chain descends as history accumulates, so it enters the last-window analysis time from a lower error than the cold single-window methods. The cycling advantage is most pronounced at $\sigma_{\mathrm{obs}} = 0.5$ with long history $T_{\mathrm{obs}} = 48$, and marginal at $\sigma_{\mathrm{obs}} = 0$. Deep in the chaotic forecast all methods converge to the same climatological saturation: cycling improves the analysis and short-lead forecast, not the long-range predictability.

lightweight defaults (200 trajectories, 30 epochs) rather than the paper-scale 32,000 trajectories and 500 epochs; and the sliding-window experiments sweep only a single noise level, $\sigma_{\mathrm{obs}} = 0.0$. All quantitative claims below should be read with these caveats foregrounded.

Turbulent trajectory and warmup diagnostic. Figure 13 shows vorticity snapshots from a representative training trajectory. The fields display the characteristic turbulent shear bands at the $k = 4$ forcing scale, with vorticity spanning roughly $[-12.25, 14.08]$; this is the turbulent state that data assimilation (DA) must reconstruct from only a 4×4 subsampled observation grid. Warmed-up initial conditions are produced by integrating a spectrally-filtered random vorticity field forward $N_{\mathrm{warmup}} = 100$ outer steps into an approximately quasi-stationary turbulent regime. Figure 14

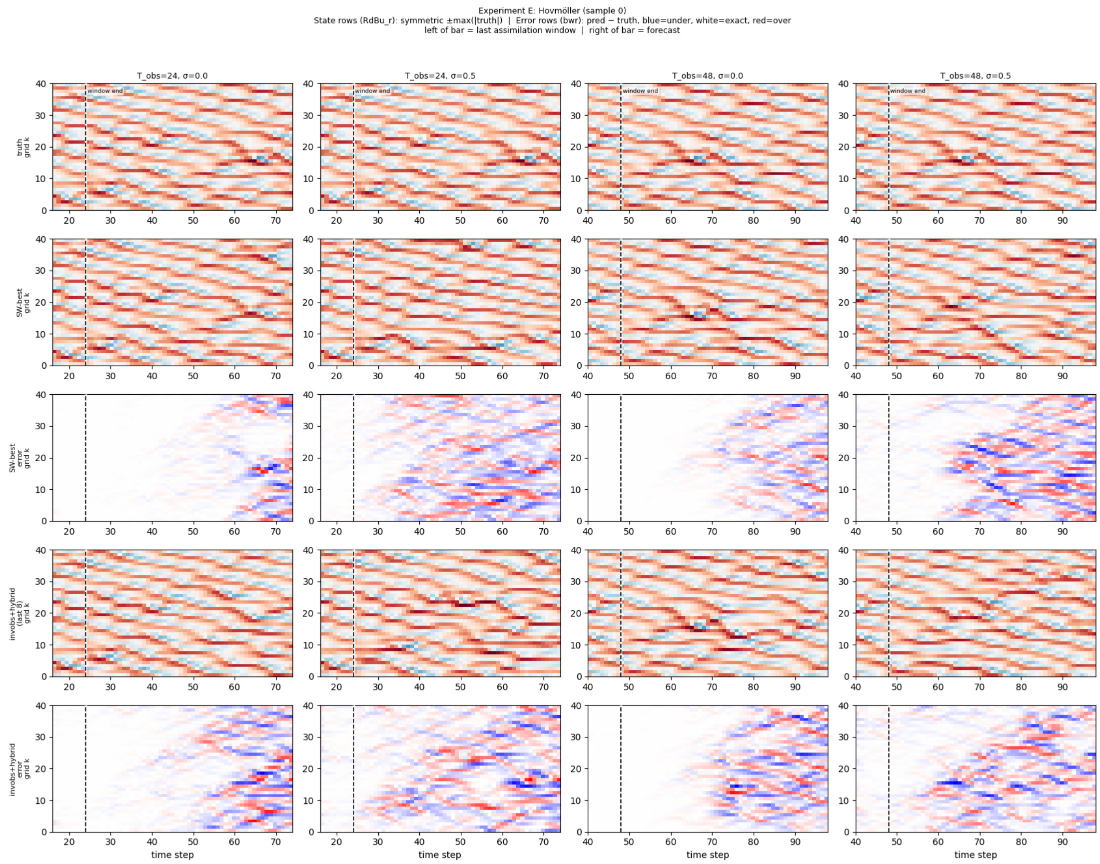

Figure 12: Experiment D: space-time (Hovmöller) diagnostics for one ensemble member (sample 0) of the Lorenz-96 sliding-window run. Columns are the four configurations $(T_{\text{obs}}, \sigma_{\text{obs}}) \in \{24, 48\} \times \{0.0, 0.5\}$; the five rows are (1) truth, (2) SW-best reconstructed state, (3) SW-best signed error (prediction $-$ truth), (4) invobs+hybrid (last-8) reconstructed state, and (5) invobs+hybrid signed error, with grid index k on the vertical axis and time step on the horizontal axis. State rows use a diverging colormap symmetric about $\pm \max |\text{truth}|$; the two error rows share a scale set to the larger $\max |\text{error}|$ across both methods within each column (blue = under-prediction, white = near-exact, red = over-prediction), making the two methods directly comparable. The dashed vertical line marks the window end T_{obs}: left of it is the last assimilation window, right of it the 50-step forecast. Both methods reproduce the truth's travelling wave packets through and just past the window; the signed-error rows are near-white inside the window and develop structured stripes as the chaotic forecast decorrelates, with SW-best visibly cleaner near the window at $\sigma_{\text{obs}} = 0.5$. This is a single illustrative member ($N = 50$ overall), not a statistical summary.

tracks the enstrophy $\frac{1}{2}\langle \omega^2 \rangle$ along a single warmup trajectory: it starts at 0.500 from the smooth cold start, overshoots to a peak near 15.3 around step 20 as small scales spin up, then relaxes into a quasi-stationary band, reading 9.824 at $t = 100$ and 6.195 at $t = 150$. The last-20-step relative drift is 10.82%, marginally above the $\lesssim 10\%$ heuristic stationarity target, so we report the flow honestly as *approximately* quasi-stationary rather than fully converged — a direct consequence of the short single-trajectory diagnostic. The accompanying snapshots evolve from a featureless filtered field into developed $k = 4$ turbulence by step 25 onward.

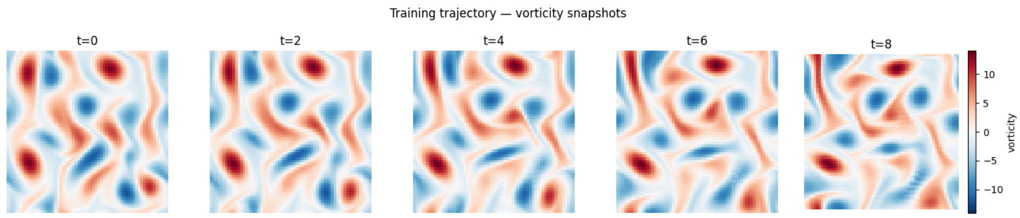

Figure 13: Vorticity snapshots of a single representative training trajectory for the 2D Kolmogorov flow on the 64×64 grid, shown at five equally spaced times along the $T = 10$-step assimilation window. Each panel plots the vorticity ω with a diverging red–blue colormap (`RdBu_r`) scaled symmetrically to $\pm \max |\omega|$. The fields exhibit the characteristic turbulent shear bands at the $k = 4$ forcing scale that the assimilation experiments must reconstruct; the vorticity range across this trajectory is approximately $[-12.25, 14.08]$. This is the turbulent, approximately quasi-stationary state being assimilated from only a 4×4 subsampled observation grid.

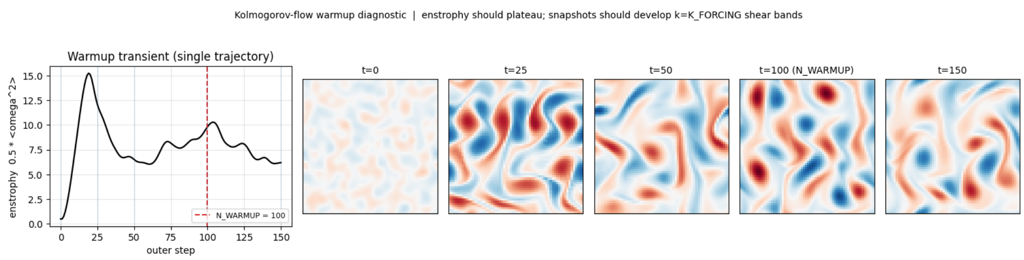

Figure 14: Warmup and stationarity diagnostic for the Kolmogorov-flow spin-up, on a single trajectory. Left: enstrophy $\frac{1}{2}\langle\omega^2\rangle$ versus outer integration step; the red dashed line marks the warmup length $N_{\text{warmup}} = 100$. Enstrophy starts at 0.500 from the smooth filtered cold start, overshoots to a peak near 15.3 around step 20, then relaxes to 9.824 at $t = 100$ and 6.195 at $t = 150$. The remaining panels show vorticity snapshots (`RdBu_r`, shared scale) at $t = 0, 25, 50, 100, 150$, evolving from a featureless field into developed $k = 4$ turbulent shear bands. The last-20-step relative drift is 10.82%, marginally above the $\lesssim 10\%$ heuristic, so the flow is reported as approximately quasi-stationary. Integrating 100 warmup steps brings the cold start into a turbulent, approximately stationary regime suitable for generating assimilation experiments.

Experiment A: single 10-step window. With $T_{\text{obs}} =$ `WINDOW_T` $= 10$, cold-start background-error standard deviation $\sigma_{\text{b}} = 1.0$, and $\sigma_{\text{obs}} = 0.0$, we run the four initialization×optimization combinations (learned-inverse “invobs” or bicubic-interpolation “baseline” initialization, paired with hybrid or observation-only optimization). The clearest discriminator is the final limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) cost: invobs initialization lands the optimizer in a far lower-loss basin than the baseline. The final (unnormalized, summed $J_{\text{b}} + J_{\text{o}}$) losses are invobs+hybrid 56,385 and invobs+obs-only 45,760 versus baseline+hybrid 168,341 and baseline+obs-only 138,254 — invobs starts the optimizer roughly $3\times$ lower in cost, at comparable iteration counts ($\approx$ 261–270). The analysis root-mean-square error (RMSE) at the window start orders the same way: the invobs combinations reach $\approx$ 3.5–3.7 while the baseline combinations sit near 4.8–5.5 (Figure 15). The spatial-RMSE-over-window curves separate clearly during assimilation, but all rollouts converge toward the saturation RMSE once the forecast leaves the window, as expected for a chaotic flow observed at only 16 of 4096 points.

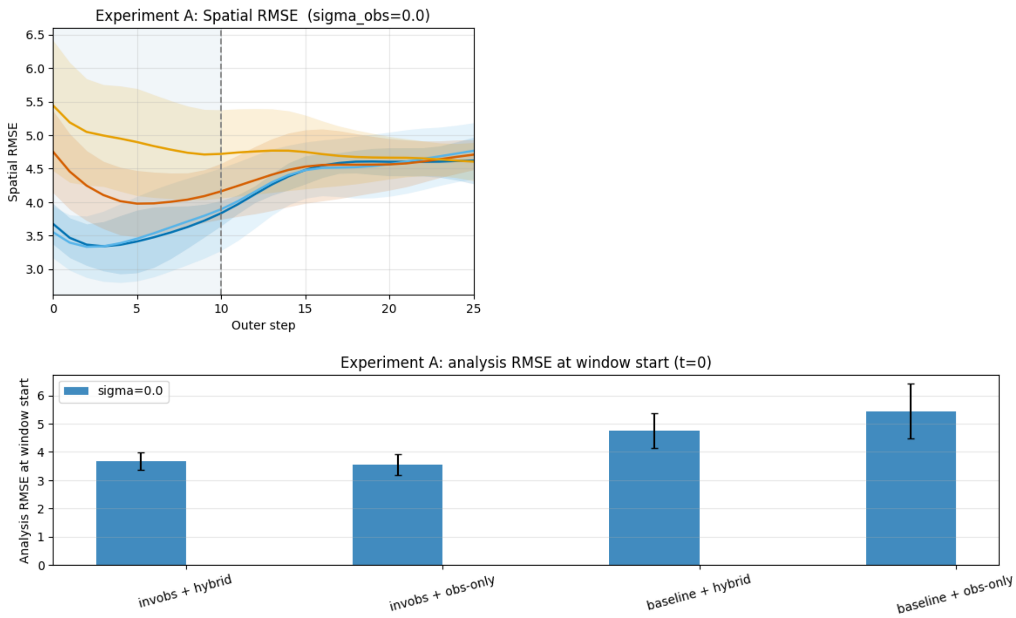

Figure 15: Kolmogorov Experiment A: a single 10-step assimilation window ($T_{\text{obs}} = \texttt{WINDOW\_T} = 10$, $\sigma_{\text{b}} = 1.0$, $\sigma_{\text{obs}} = 0.0$), comparing the four initialization×optimization combinations. Top: ensemble-mean spatial RMSE (± 1 standard deviation over the $N = 8$ ensemble) versus outer step, spanning the assimilation window (shaded, 0–10, with the dashed window-end line) and the free forecast to step 25. Bottom: bar chart of analysis RMSE at the window start ($t = 0$). The invobs combinations reach lower in-window RMSE (≈ 3.5–3.7) than the baseline combinations (≈ 4.8–5.5), but all curves converge toward saturation once the forecast leaves the window, reflecting the chaotic flow and extreme observation sparsity (16 of 4096 points). The final L-BFGS costs (invobs+hybrid 56,385, invobs+obs-only 45,760 versus baseline+hybrid 168,341, baseline+obs-only 138,254) show invobs initialization places the optimizer in a basin roughly 3× lower in cost.

Experiment B: stride sweep. We next run sliding-window (cycling) 4D-Var with invobs initialization every cycle, hybrid optimization, the propagated previous analysis as the J_{b} background on cycles ≥ 1, and a tighter cycling background-error standard deviation $\sigma_{\text{b}} = 0.3$. Figure 16 sweeps strides $\{2, 5, 10\}$ for total observation horizons $T_{\text{obs}} \in \{20, 30\}$ at $\sigma_{\text{obs}} = 0.0$. The stride curves overlap heavily within the shaded last-assimilation window and rise to the same saturation RMSE (≈ 4.6–4.8) once the forecast begins; the coarsest stride, 10, gives the lowest mean forecast RMSE for *both* horizons, and the best single-window reference method is invobs+hybrid. The dashed best-Experiment-A reference, shifted to the last-window position, tracks the cycling curves closely, indicating that the cycling machinery does not yet separate from a single well-initialized window at this smoke scale.

Experiment C: cycling versus single-window. Figure 17 is the central comparison: the best-stride sliding-window run (full per-cycle analysis chain) against single-window invobs+hybrid and single-window baseline+obs-only, the latter two applied only to the last 10 observations. The sliding-window-best curve traces per-cycle analysis RMSE from $t = 0$ as a dashed marked chain and hands off to a solid curve over the last cycle plus its forecast. All three methods converge to the same saturation band (≈ 4.6) once the forecast leaves the window. The honest reading of this smoke-scale run is a negative one: with $N = 8$ and a single noise level, cycling does not deliver a

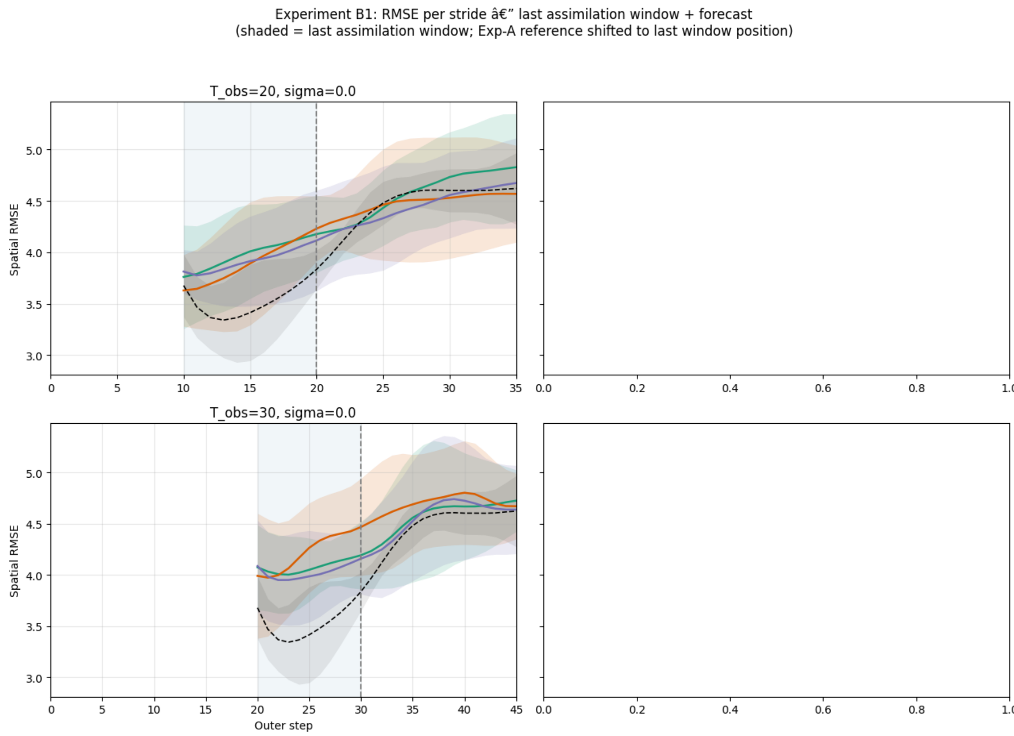

Figure 16: Kolmogorov Experiment B1: sliding-window (cycling) 4D-Var stride sweep at $\sigma_{\text{obs}} = 0.0$, with invobs initialization every cycle, hybrid optimization, the propagated previous analysis as the J_{b} background on cycles ≥ 1, and cycling $\sigma_{\text{b}} = 0.3$. Rows are observation horizons $T_{\text{obs}} = 20$ (top) and 30 (bottom); only the $\sigma_{\text{obs}} = 0.0$ column is populated (the right column of the 2×2 layout is empty because only one noise level was swept). Each panel plots ensemble-mean spatial RMSE (± 1 standard deviation) versus outer step for strides 2, 5, and 10 (coloured), over the last assimilation window (shaded, ending at the dashed line at T_{obs}) and the forecast to $T_{\text{obs}} + 15$; the black dashed curve is the best single-window reference (invobs+hybrid) shifted to the last-window position. The stride curves overlap heavily and rise to the same saturation RMSE (≈ 4.6–4.8); the coarsest stride 10 gives the lowest mean forecast RMSE. At this smoke scale ($N = 8$, one noise level) cycling does not yet separate from the single-window reference.

decisive forecast advantage over a single well-initialized invobs+hybrid window once predictability is lost. (This contrasts with the Lorenz-96 cycling results of Section 4.1, where a noise- and history-dependent cycling benefit was visible at $\sigma_{\text{obs}} = 0.5$; the Kolmogorov runs sweep only $\sigma_{\text{obs}} = 0$, the regime where cycling helps least, so the comparison here is not apples-to-apples.)

Vorticity snapshot comparison. Finally, Figure 18 gives a qualitative 5×3 view of the reconstructed fields for Experiment C at $T_{\text{obs}} = 20$ (ensemble sample 0). The rows are ground-truth vorticity, the sliding-window-best prediction and its signed error, and the single-window invobs+hybrid prediction and its signed error; the columns are window start, window end, and forecast end. At window start, both methods reproduce the large-scale vorticity structures with small error. By forecast end, the error fields grow to the magnitude of the state itself, visually corroborating the RMSE saturation seen in the quantitative panels and underscoring the fundamental difficulty of constraining a turbulent 64×64 field from only 16 point observations.

In summary, the Kolmogorov-flow runs reproduce the paper's central mechanism — learned-

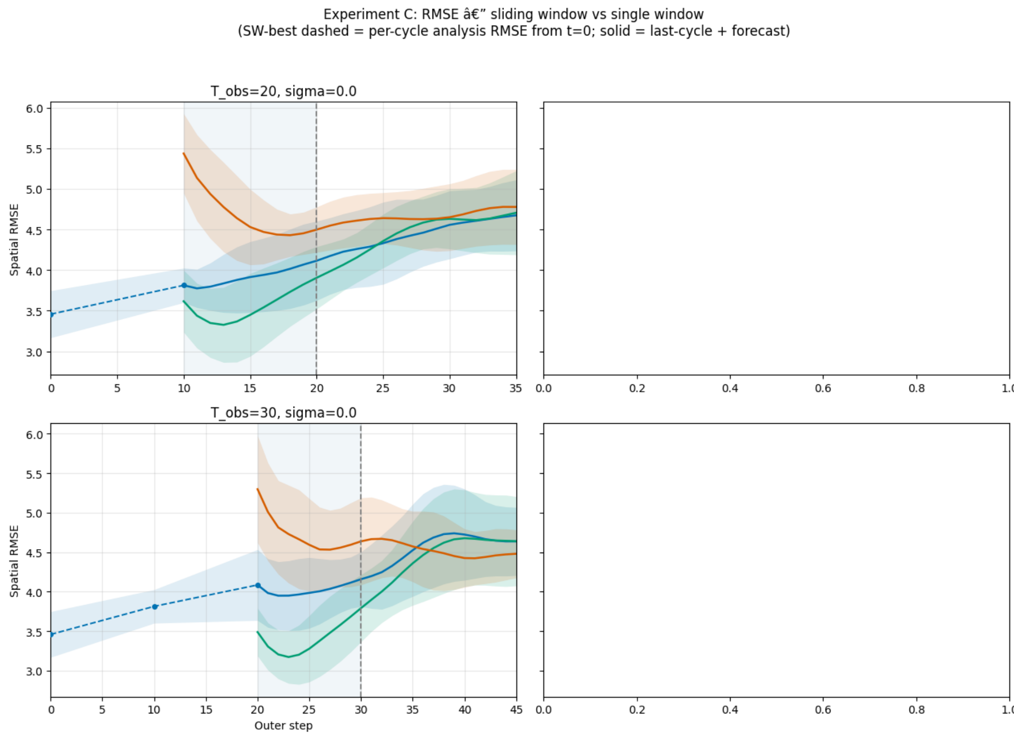

Figure 17: Kolmogorov Experiment C: the best-stride sliding-window run versus single-window baselines at $\sigma_{\text{obs}} = 0.0$. Rows are $T_{\text{obs}} = 20$ (top) and 30 (bottom); only the $\sigma_{\text{obs}} = 0.0$ column is populated. Each panel plots ensemble-mean spatial RMSE (± 1 standard deviation) versus outer step. The sliding-window-best result is a dashed marked curve of per-cycle analysis RMSE from $t = 0$, continued as a solid curve over the last cycle plus its forecast; it is compared against single-window invobs+hybrid and single-window baseline+obs-only, each applied to the last 10 observations (shaded last-window region, dashed window-end line at T_{obs}, forecast to $T_{\text{obs}} + 15$). All three methods converge to the same saturation band (≈ 4.6) once the forecast leaves the window: at this smoke scale cycling delivers no decisive advantage over a single well-initialized invobs+hybrid window.

inverse initialization places 4D-Var in a markedly lower-loss basin (Experiment A) and produces the best in-window analyses — but at this CPU-limited, $N = 8$, single-noise-level smoke scale the operational cycling extension does not yet demonstrate a clear forecast advantage, and all methods saturate to climatological error once the chaotic flow decorrelates from truth. These outcomes are best understood as a working demonstration of the pipeline on a hard 2D system rather than as converged comparisons.

5 Discussion and Conclusion

5.1 Summary of findings

Our experiments tested whether the learned inverse-observation idea of Frerix et al. [1] survives a more operationally realistic setting: the full four-dimensional variational data assimilation (4D-Var) cost $J = J_{\text{b}} + J_{\text{o}}$ with an explicit background term and observation-error covariance, Gaussian observation noise on every measurement, and an operational sliding-window (cycling) extension. The

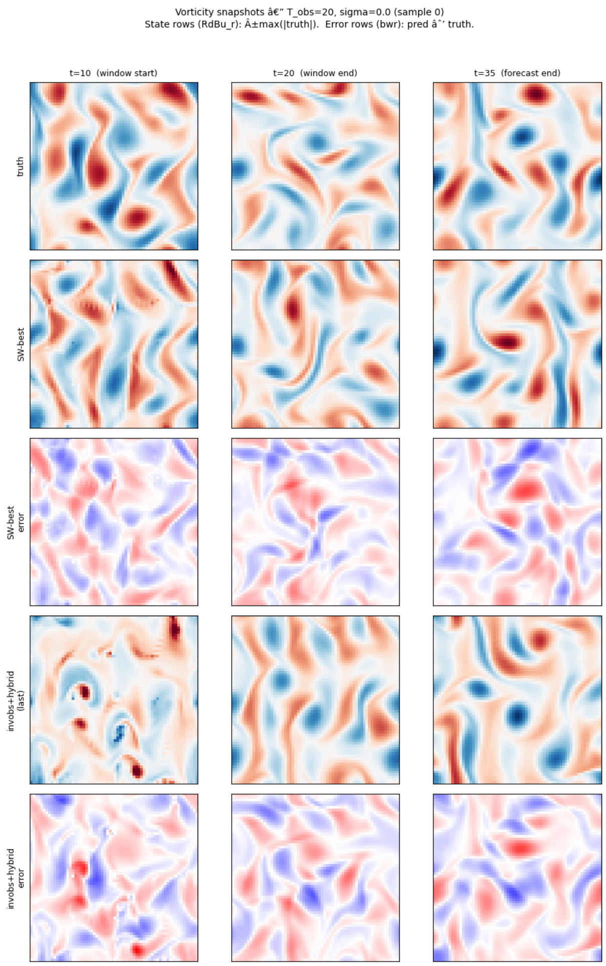

Figure 18: Kolmogorov vorticity-field snapshot comparison for Experiment C at $T_{\text{obs}} = 20$, $\sigma_{\text{obs}} = 0.0$ (ensemble sample 0). The 5×3 grid has columns at three times relative to the last-window start: window start ($t = 10$), window end ($t = 20$), and forecast end ($t = 35$). Rows top to bottom: ground-truth vorticity; the sliding-window-best prediction; the sliding-window-best signed error; the single-window invobs+hybrid prediction; and the invobs+hybrid signed error. State rows use a diverging red–blue colormap (`RdBu_r`) scaled to $\pm \max |\text{truth}|$; the two error rows share a single symmetric blue–white–red scale (`bwr`) showing prediction minus truth. At window start both methods reproduce the large-scale structures with small error, but by forecast end the error fields grow to the magnitude of the state itself, visually corroborating the RMSE saturation and the difficulty of reconstructing a 64×64 turbulent field from 16 point observations.

central claim of the paper broadly held. On Lorenz-96, under the tuned-σ_{b} full 4D-Var cost, learned-inverse-observation initialization combined with hybrid (physics-space then observation-space) optimization attained the lowest analysis root-mean-square error (RMSE) at every tested noise level (RMSE $= 0.620$, 1.168, and 1.676 at $\sigma_{\text{obs}} = 0.1$, 0.5, and 1.0 respectively). Equally important, hybrid optimization rescued the cheap climatological-mean baseline from the bad observation-space local minima that trap pure observation-space 4D-Var when only 10 of 40 components are observed: it roughly halved the analysis error (for example $2.393 \to 1.223$ at $\sigma_{\text{obs}} = 0.5$). The same mechanism appeared in the sliding-window experiments, where baseline initialization with observation-only optimization remained stuck near the climatological RMSE while every hybrid-warm-started method reached an analysis RMSE near 1 at $\sigma_{\rm obs} = 0$. Cycling delivered a real but conditional benefit: the per-cycle analysis dropped below the single-window reference once history accumulated, and the advantage was clearest at higher noise and longer observation horizons ($\sigma_{\rm obs} = 0.5$, $T_{\rm obs} = 48$), while being marginal at $\sigma_{\rm obs} = 0$, where a single well-posed window already over-determines the observed components. On 2D Kolmogorov flow the qualitative story carried over: learned-inverse-observation initialization placed L-BFGS in a basin roughly 3× lower in cost than bicubic interpolation (final losses 56,385 and 45,760 for the invobs combinations versus 168,341 and 138,254 for the baseline), and produced the best in-window analyses. In every case, once the chaotic forecast left the assimilation window all methods relaxed to the same climatological saturation, consistent with the short measured Lyapunov time of the system—analysis quality improves the short-lead forecast, not the Lyapunov-limited long-range predictability.

5.2 Caveats and limitations

This is a course capstone, and several limitations should temper how strongly its conclusions are read.

Small ensembles. The tuned-$\sigma_{\rm b}$ Lorenz-96 sweep used only $N = 16$ trajectories, the sliding-window runs used $N = 50$, and the Kolmogorov sliding-window runs used only $N = 8$. Consequently several "best" selections—the best stride, the best single-window method at $\sigma_{\rm obs} = 0$, and the cycling-versus-single-window margin in the clean-observation cases—fall within ensemble scatter and should be read as tendencies rather than decisive separations.

Compute-limited Kolmogorov runs. The Kolmogorov solver relies on `torch.fft.rfft2`, which ran on CPU only in our stack, so all 2D experiments were validation-scale: a tiny ensemble, a lightweight inverter trained on only a few hundred trajectories for a few tens of epochs rather than the paper-scale 32,000 trajectories and 500 epochs, and a single observation-noise level. These runs demonstrate the pipeline mechanics and reproduce the qualitative basin-of-attraction benefit, but they are not converged scientific results, and the cycling machinery did not yet separate from a single well-initialized window at this scale. Relatedly, the warmup diagnostic reached only approximate quasi-stationarity (a last-20-step enstrophy drift of about 10.8%, marginally above the $\lesssim 10\%$ heuristic), a direct artifact of the short single-trajectory diagnostic.

Inverter trained at one noise level. The learned inverse observation operator H_θ^{-1} is supervised at a fixed noise level and then, in some cells, evaluated under a deliberate mismatch. This probes robustness but means the inverter is never optimal for the noise it actually encounters; a per-noise checkpoint was supplied only where feasible.

Short measured Lyapunov time. Our fixed-step fourth-order Runge–Kutta (RK4) integrator yields a fitted Lyapunov time $T_L = 0.442$ model time units (MTU), shorter than the paper's expected ~ 0.6 MTU. We report this honestly as a known consequence of the coarse fixed step being slightly too coarse to resolve the fastest unstable directions. It sets the chaotic saturation rate to which every method relaxes in the forecast and slightly tightens the window over which our RK4 and an adaptive Dormand–Prince (dopri5) integrator represent the same physics (median crossover $t^\star = 7.70$ MTU); our assimilation windows ($T = 1.0$ MTU) sit comfortably inside that horizon, so the integrator choice does not contaminate the data assimilation (DA) results.

The negative $\sigma_{\rm b}$-ordering result. We expected invobs+hybrid to prefer a *smaller* optimal background-error standard deviation $\sigma_{\rm b}$ than the paper baseline—reasoning that the learned lift supplies a stronger physics-space prior, so the analysis could lean on the background more (a smaller $\sigma_{\rm b}$ raises the background weight $1/(2\sigma_{\rm b}^2)$). This did not hold: invobs+hybrid selected $\sigma_{\rm b} = 10.0/1.0/1.0$ while paper+obs selected $\sigma_{\rm b} = 0.1/0.3/0.3$, the opposite direction at all three noise levels. Moreover, tuning σ_b helped only marginally over the untuned $\sigma_\text{b} = 1$ in most cells (gains typically below 0.2 RMSE, and exactly zero where the best σ_b was already 1.0).

No real data and simplified covariances. All experiments are on synthetic, perfect-model trajectories with no model error, and our covariances are deliberately simple: a single static spatial-correlation matrix $\boldsymbol{C}$ (condition number 8.72) for the Lorenz-96 background $\boldsymbol{B} = \sigma_\text{b}^2\boldsymbol{C}$, a diagonal observation-error covariance $\boldsymbol{R} = \sigma_\text{obs}^2\boldsymbol{I}$ with independent noise, and for Kolmogorov flow an identity background $\boldsymbol{B} = \sigma_\text{b}^2\boldsymbol{I}$ because forming and factorizing a full 4096×4096 spatial covariance was prohibitively expensive. None of these capture the flow-dependent, correlated error structures of an operational system.

5.3 Future work

Each limitation suggests a concrete next step. The inverter could be made *noise-aware*—either a family of per-noise checkpoints or a single network conditioned on σ_obs—so that H_θ^{-1} is always matched to the observation error it encounters. The Kolmogorov pipeline should be moved to a GPU and rerun at paper scale, with much larger ensembles, a fully trained inverter, and a sweep over multiple noise levels, so that the cycling extension can be evaluated where it has a chance to separate from a single window. On the statistical side, both covariances invite enrichment: a *correlated* observation-error covariance $\boldsymbol{R}$ in place of the diagonal $\sigma_\text{obs}^2\boldsymbol{I}$, and a flow-dependent background covariance estimated from an ensemble (an ensemble or hybrid ensemble-variational scheme) in place of the static $\boldsymbol{C}$, which is also the natural route to a tractable background covariance for the 2D flow without forming the full dense matrix. Finally, an adaptive integrator (dopri5) would produce paper-matched trajectories with the expected Lyapunov time, removing the fixed-step artifact and enabling a cleaner quantitative comparison against Frerix et al. [1]; larger ensembles throughout would turn the present tendencies into statistically firm claims.

5.4 Conclusion

Across both a 1D chaotic toy model and a 2D turbulent flow, and under a full 4D-Var cost with noisy observations that the original study did not consider, the learned inverse observation operator continued to help where it was designed to: it provides a physics-space initialization and a near-convex hybrid objective that together place gradient-based 4D-Var in a good basin and yield the lowest analysis error, especially when sparse, lossy observations would otherwise trap pure observation-space optimization. The operational sliding-window extension showed that cycling can further reduce analysis error as history accumulates, most clearly when observations are noisy. At the same time we report honest negatives—an unmet expectation about the optimal background-error standard deviation, a short integrator-limited Lyapunov time, and compute-limited 2D runs—that mark exactly where larger ensembles, GPU-scale Kolmogorov experiments, richer covariances, and an adaptive integrator would turn these encouraging tendencies into firm conclusions.

References

[1] Thomas Frerix, Dmitrii Kochkov, Jamie A. Smith, Daniel Cremers, Michael P. Brenner, and Stephan Hoyer. Variational data assimilation with a learned inverse observation operator, 2021. Proceedings of the 38th International Conference on Machine Learning (ICML).